

Fitbuddy AI Fitness Plan Generator

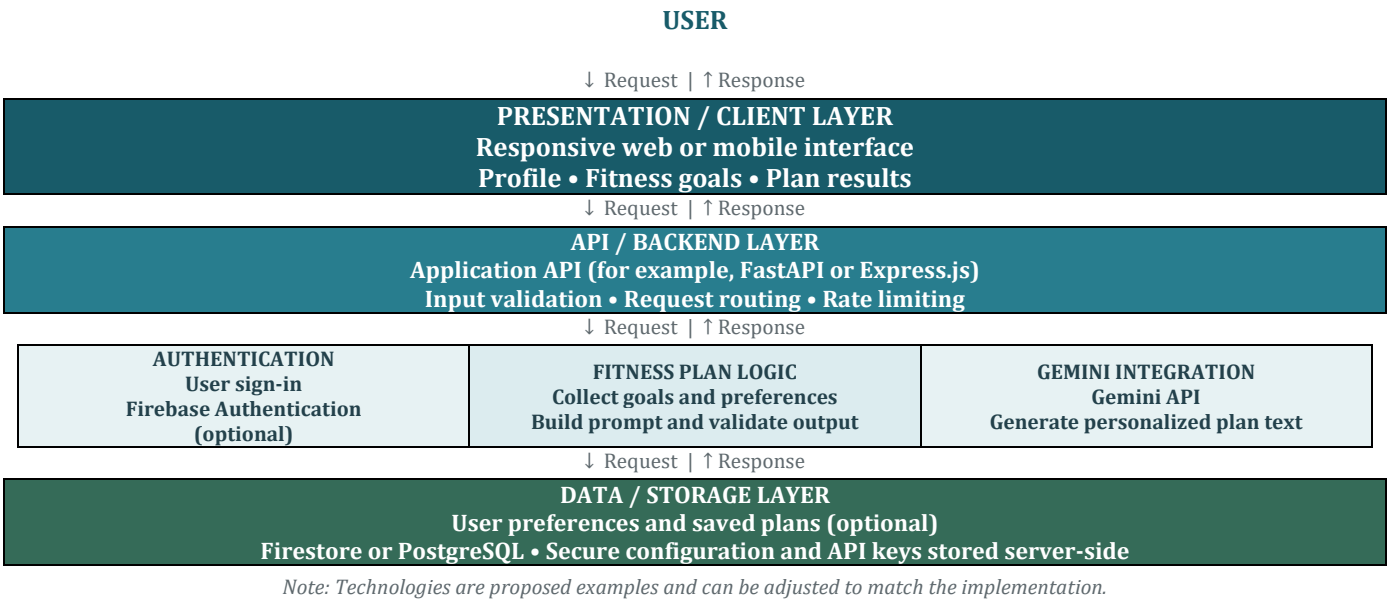
Powered by Google Gemini Models

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1. Project Overview

Fitbuddy AI Fitness Plan Generator is a proposed application that creates personalized fitness-plan suggestions using user-provided goals, fitness level, available equipment, schedule, and preferences. Google Gemini models are used to generate plan recommendations. The application should present results as general wellness guidance and remind users to consult a qualified professional when they have medical conditions, injuries, or other health concerns.

2. Solution Architecture Diagram



3. Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Allows users to enter fitness goals, experience level, schedule, equipment, and preferences; displays generated plans.	HTML/CSS/JavaScript or React; responsive UI
API / Backend	Receives requests, validates inputs, coordinates application logic, and returns the generated plan to the client.	FastAPI (Python) or Express.js (Node.js)
Authentication Service	Optionally manages sign-in and user identity before saving personal plans or profile information.	Firebase Authentication or OAuth 2.0
Fitness Plan Logic	Converts user details into a structured prompt, checks the returned content, and formats a readable plan.	Python or JavaScript; validation rules
External AI Integration	Sends the prompt to a Gemini model and receives generated fitness-plan suggestions.	Google Gemini API / Gemini models
Database / Storage	Optionally stores user preferences and saved plans. Store only information needed by the app and restrict access.	Cloud Firestore or PostgreSQL
Security & Configuration	Keeps API keys on the server, applies access controls, and avoids exposing secrets in the client application.	Environment variables; HTTPS; server-side access controls

4. Basic Request Flow

1. The user enters fitness goals, fitness level, preferred workout duration, equipment, and schedule.
2. The client sends the information to the backend API over HTTPS.
3. The backend validates the input and prepares a prompt for the Gemini API.
4. Gemini returns a draft plan; the backend checks and formats the response before returning it.
5. The app displays the plan. If enabled, the user can save it to the database.

5. Safety, Privacy, and Assumptions

- Fitness plans are general informational suggestions, not medical diagnosis or treatment.
- Ask users to consult a qualified healthcare or fitness professional when appropriate, especially for injuries or medical conditions.
- Do not send unnecessary personal or sensitive information to the AI service.
- Keep the Gemini API key on the server; never embed it in browser or mobile client code.
- Authentication and database storage are optional design choices and should be enabled only if required.

Prepared for the Fitbuddy AI project team